

Full Stack Engineering

Project Report

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Social Media Platform

CHITKARA
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Abstract

The TrueVoice platform is a full-stack social media web application designed to enhance digital interaction while ensuring content authenticity and user safety. In existing social platforms, issues such as content theft, lack of proper attribution, and delayed response to online harassment create challenges for users.

This system provides a centralized platform where users can create and share posts, interact through likes and comments, follow other users, and receive real-time notifications. It introduces role-based access control and secure authentication mechanisms to ensure safe and controlled user interactions.

A key feature of the platform is **content fingerprinting using SHA-256 hashing**, which detects duplicate content and ensures that original creators receive proper credit. Additionally, an intelligent **shield system monitors comment activity in real time** and restricts abusive or excessive interactions, improving user safety.

The application is developed using Node.js and Express.js for backend services, React.js for frontend development, and MySQL as the database. Real-time features are implemented using Socket.IO, and RESTful APIs are used for efficient communication between system components.

The proposed system aims to provide a secure, scalable, and user-friendly social media experience. It can be further enhanced with advanced features such as content recommendations, improved moderation techniques, and mobile application support.

1. Introduction :

In today's fast-paced digital world, social media platforms play a vital role in enabling communication, content sharing, and user interaction. However, existing platforms face significant challenges such as content theft, lack of proper attribution, and delayed handling of online harassment. These issues often lead to reduced trust, poor user experience, and unsafe online environments.

A modern social media platform must not only support seamless interaction but also ensure content authenticity and user safety. TrueVoice is designed as a comprehensive solution that provides a structured and secure environment where users can create, share, and interact with content in real time. It offers a centralized platform for posting content, engaging through likes and comments, following other users, and receiving real-time notifications.

The proposed TrueVoice system aims to address key challenges present in traditional social media platforms. It introduces content fingerprinting to detect duplicate content and ensure proper credit to original creators. Additionally, it implements an intelligent shield system that monitors comment activity and prevents abusive or excessive interactions. Users can manage their profiles, interact safely, and stay informed through notifications, while administrators can monitor system activities effectively.

This project is developed using modern technologies such as Node.js and Express.js for backend development, React.js for frontend design, and MySQL for database management. Real-time features are implemented using Socket.IO, and RESTful APIs are used for efficient communication between system components.

Overall, TrueVoice provides a secure, scalable, and user-centric approach to social media interaction, enhancing content integrity, improving user safety, and delivering a better digital communication experience.

1.1 Background

In recent years, the rapid growth of social media platforms has transformed the way people communicate, share content, and interact online. However, with this growth, several challenges have emerged, including content theft, lack of proper attribution, and increasing instances of online harassment. While traditional social platforms provide basic interaction features, they often struggle to ensure content authenticity and user safety at scale.

With the advancement of technology and digital ecosystems, there is a growing need for platforms that go beyond simple content sharing and focus on trust, transparency, and protection. Modern web applications are expected to provide real-time interaction, secure communication, and intelligent moderation systems. These requirements have led to the development of more advanced and structured social media solutions.

Contemporary platforms aim to improve user engagement and safety through features such as real-time updates, notifications, and personalized content feeds. However, proactive mechanisms for handling duplicate content and preventing abusive interactions are still limited. This creates a gap where users may face issues like uncredited content reuse and overwhelming negative interactions.

The development of web technologies such as Node.js, Express.js, and relational databases like MySQL has enabled the creation of scalable and efficient applications. Additionally, real-time communication tools like Socket.IO and secure authentication mechanisms such as JWT have made it possible to build responsive and secure platforms.

The proposed TrueVoice system is developed in response to these challenges and technological advancements. It aims to provide a secure and intelligent social media platform by integrating content fingerprinting for originality protection and a real-time shield system for preventing harassment, thereby enhancing user trust, safety, and overall experience.

1.2 Objective

The main objective of the TrueVoice system is to develop a secure and user-friendly social media platform that enables users to share content, interact in real time, and ensure content authenticity and user safety.

The specific objectives of the project are as follows:

- To design and develop a centralized platform for content sharing and user interaction.
- To allow users to create, manage, and share posts with text and media.
- To enable users to follow others and view a personalized news feed.
- To provide real-time interaction through likes, comments, and notifications.
- To implement content fingerprinting to detect duplicate content and ensure credit to original creators.
- To develop a shield system to monitor and control abusive or excessive comments.
- To allow users to manage their profiles and account settings efficiently.
- To generate real-time notifications for user activities such as likes, comments, and follows.
- To maintain secure authentication and authorization using JWT.
- To ensure efficient data management using MySQL database.
- To improve user safety, content integrity, and overall platform experience.

1.3 Significance

The TrueVoice platform holds significant importance in improving the way users interact and share content on digital platforms. In traditional social media environments, issues such as content theft, lack of proper attribution, and delayed handling of online harassment often lead to poor user experience and reduced trust. This project addresses these challenges by providing a secure and structured platform for content sharing and user interaction.

One of the key significances of this system is that it enhances content integrity by ensuring that original creators receive proper credit through content fingerprinting. Users can confidently share their work without the risk of uncredited duplication. The platform also promotes better interaction by enabling users to engage through likes, comments, and real-time notifications.

Another important aspect is improved user safety and moderation. With the implementation of an intelligent shield system, the platform can monitor comment activity and restrict abusive or excessive interactions. This proactive approach helps create a safer and more controlled environment for users.

Additionally, the system ensures transparency and accountability through activity tracking and secure authentication mechanisms. Role-based access and proper authorization enhance data security and ensure controlled access to system functionalities.

Overall, TrueVoice contributes to a more secure, reliable, and user-centric social media experience by improving content protection, enhancing user safety, and enabling efficient real-time interaction, making it a valuable solution for modern digital communication platforms.

2. Problem Definition And Requirement :

In modern social media platforms, users face several challenges related to content sharing and interaction. Issues such as content theft, lack of proper attribution, and delayed response to online harassment reduce user trust and negatively impact the overall experience. Traditional platforms focus mainly on engagement but often lack mechanisms to ensure content authenticity and user safety.

The major problems identified are the absence of content ownership protection, ineffective handling of abusive or excessive interactions, lack of real-time monitoring systems, and limited control over user-generated content. Additionally, there is no efficient system to track content origin or maintain transparency in user interactions.

Therefore, there is a need for a centralized and intelligent platform that can enable secure content sharing, ensure proper credit to original creators, monitor interactions in real time, and provide a safe and reliable user experience.

1. Functional Requirements

The functional requirements describe the core features and operations of the system:

- The system should allow users to register and log in securely.
- The system should allow users to create, edit, and delete posts with text and media.
- The system should allow users to like, comment, and interact with posts.
- The system should allow users to follow and unfollow other users.
- The system should provide a personalized news feed based on user activity.
- The system should implement content fingerprinting to detect duplicate content.
- The system should ensure proper credit attribution for original content creators.
- The system should monitor comments and control abusive interactions using a shield mechanism.
- The system should generate real-time notifications for likes, comments, and follows.
- The system should allow users to manage their profiles and account settings.
- The system should maintain activity logs for tracking user actions.

2. Non-Functional Requirements

The non-functional requirements define the quality attributes of the system:

- **Performance:** The system should respond quickly and support real-time interactions efficiently.
- **Security:** The system should implement secure authentication (JWT) and protect user data.
- **Scalability:** The system should handle a growing number of users and content without performance issues.
- **Reliability:** The system should function consistently and maintain data integrity.
- **Usability:** The system should provide a simple, intuitive, and user-friendly interface.
- **Maintainability:** The system should be modular and easy to update or extend in the future.

3. References

For making our website we took a help of following sites-

- http://www.w3schools.com/css/css_background.asp
- http://www.w3schools.com/js/js_datatypes.asp
- http://www.w3schools.com/sql/sql_insert.asp
- http://www.w3schools.com/sql/sql_update.asp